WORKED SOLUTION

STEP II 2008, Question 10

Pure • Collision

STEP II 2008, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s2-q10

The key idea

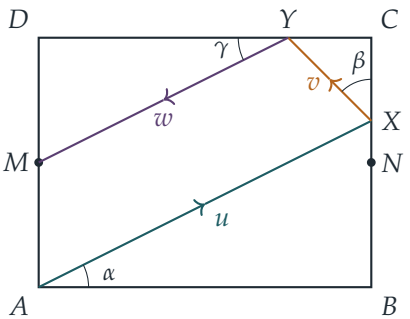
At a smooth wall the impulse is perpendicular to the wall, so the velocity component along the wall is unchanged and only the component into the wall is multiplied by e. Splitting the velocity into components parallel to AB and to AD makes everything transparent: each bounce scales just one of the two components. After both bounces, both components have been scaled by e, which settles the final direction and the kinetic energy at once. The geometry of part (ii) is then a matter of adding up horizontal distances.

Solution

We model the ball as a particle and the sides as smooth walls. Take axes with origin A, the x-axis along AB and the y-axis along AD, so that

$$B = (a, 0), \quad C = (a, 2b), \quad D = (0, 2b), \quad M = (0, b), \quad N = (a, b).$$

Write $t = \tan\alpha$ for short.



Part (i)

Initially the velocity is $(u\cos\alpha,\ u\sin\alpha)$.

Impact at X. The side BC is parallel to the y-axis. Since it is smooth, the impulse on the ball is along the x-direction, so the y-component $u\sin\alpha$ is unchanged. By Newton's law of restitution the x-component is reversed and multiplied by e. The velocity after the impact is

$$(-eu\cos\alpha,\ u\sin\alpha).$$

The angle $\beta = \angle CXY$ is measured from XC, which points in the positive y-direction. So

$$\tan\beta = \frac{eu\cos\alpha}{u\sin\alpha} = \frac{e}{\tan\alpha}, \qquad \text{that is,} \qquad \tan\alpha\tan\beta = e,$$

as required.

Impact at Y. The side DC is parallel to the x-axis, so now the x-component is unchanged and the y-component is reversed and multiplied by e. The velocity after the impact is

$$(-eu\cos\alpha,\ -eu\sin\alpha) = -e\,(u\cos\alpha,\ u\sin\alpha).$$

This is a negative multiple of the initial velocity, so the final stage of the path is parallel to AX and travels in the opposite direction. The angle $\gamma = \angle DYM$ is measured from YD, which points in the negative x-direction, and the direction $(-\cos\alpha, -\sin\alpha)$ makes angle α with it. Hence

$$\gamma = \alpha.$$

Part (ii)

We follow the x-coordinate along the path.

From A to X. The ball rises $a\tan\alpha$ while crossing the width a, so $X = (a,\ at)$.

From X to Y. The ball rises a further $2b - at$, and on this stage it moves $\tan\beta$ units in x for each unit it rises (because β is measured from the vertical side). With $\tan\beta = e/t$ from part (i), the ball moves left a distance $(2b - at)\,e/t$, so

$$x_Y = a - \frac{e(2b - at)}{t}.$$

From Y to M. The ball drops from height $2b$ to height b, a drop of b, at angle $\gamma = \alpha$ below the horizontal. So the horizontal distance covered is $b/\tan\alpha$, and this must take it exactly to $x = 0$:

$$x_Y = \frac{b}{t}.$$

Ask yourself Why can we get away with tracking only the x-coordinate?

Each straight stage starts where the last one ended, and its direction is known from part (i). Once we know how far the ball rises or drops on a stage, the angle fixes how far it moves sideways. The heights at the ends of the stages are forced (0, then $2b$ at DC, then b at M), so the only condition left is that the sideways distances add up.

Equating the two expressions for x_Y and multiplying by t:

$$\begin{aligned} b &= at - e(2b - at) \\ &= at(1+e) - 2be, \end{aligned}$$

so $at(1+e) = b(1+2e)$, which gives

$$\tan\alpha = \frac{(1+2e)b}{(1+e)a},$$

as required.

The shot is always possible. This value of $\tan\alpha$ is positive for every $e > 0$, so it defines an angle α in $(0, \frac{\pi}{2})$. We still need the bounce points to land on the sides they are meant to.

- X is at height $at = \dfrac{1+2e}{1+e}\,b$. Since $1+e < 1+2e < 2+2e$ for $e > 0$, this lies strictly between b and $2b$. So X is genuinely on BC, below C, and above the pocket N, so the ball doesn't drop into N on the way.
- Y is at $x_Y = b/t = \dfrac{1+e}{1+2e}\,a$, which lies strictly between 0 and a for $e > 0$. So Y is genuinely on DC, between D and C.

The final stage then runs from Y to M in a straight line inside the table. Every stage of the path is therefore possible, whatever the value of e.

Part (iii)

From part (i), the final velocity is $-e$ times the initial velocity, so $w = eu$. (We can also see this directly: each component has been multiplied by e exactly once.) The kinetic energy falls from $\frac{1}{2}mu^2$ to $\frac{1}{2}me^2u^2$, where m is the mass of the ball. The fraction lost is therefore

$$\frac{\frac{1}{2}mu^2 - \frac{1}{2}me^2u^2}{\frac{1}{2}mu^2} = 1 - e^2.$$

This doesn't depend on a, b or the angles. (For $e > 1$ the "loss" is negative: the walls would be adding energy. That isn't physical, but the algebra holds.)

Checks

- With $a = 4$, $b = 1.5$ and $e = \frac{1}{2}$ (the diagram), the formula gives $\tan\alpha = \frac{1}{2}$. Then $X = (4, 2)$, $Y = (3, 3)$, and the line from Y with gradient $\frac{1}{2}$ drops 1.5 over a horizontal run of 3, landing exactly on $M = (0, 1.5)$. The kinetic energy lost is $1 - \frac{1}{4} = \frac{3}{4}$.
- When $e = 1$ (perfectly elastic), $\tan\alpha = 3b/(2a)$. Unfold the table by reflecting it in BC and then in DC: the path becomes a straight line from A to the image of M, which is $2a$ across and $3b$ up. That gives the same gradient, $3b/(2a)$. No energy is lost, and $1 - e^2 = 0$ agrees.
- A simulation of the bounces for 20,000 random choices of a, b, e and u confirmed $\tan\alpha\tan\beta = e$, $\gamma = \alpha$, arrival at M and the fraction $1 - e^2$, all to within 10^{-13}.

Common slips

- Applying e to the whole speed at an impact. Only the component perpendicular to the wall changes. The component along a smooth wall is untouched.
- Measuring β from BC's normal instead of from XC, which turns $\tan\beta$ into its reciprocal.
- Forgetting to check that X is above N and Y is between D and C. That check is the content of "the shot is possible".

Beyond STEP

For $e = 1$, "unfolding" a billiard path by reflecting the table turns it into a straight line. This is the starting point of mathematical billiards, which links to dynamical systems and, for rational tables, to geometry on surfaces. With $e < 1$ the reflection is no longer mirror-like, but splitting the velocity into components still makes it easy to handle. For the mechanics of impulses and restitution, see Kleppner and Kolenkow, *An Introduction to Mechanics*. For more problems of this flavour, try Siklos, *Advanced Problems in Mathematics*.

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